Project: Premier Hotel Network and IoT Automation System Using Cisco Packet Tracer

1. Scenario

Relevant zone(s):

Zones covered: Guest floors (three), Reception, Accounts Department, Server Room (web, e-mail, DNS, DHCP and IoT services).

The scenario is **Premier Hotel**, a multi-floor hotel modelled as a smart building in Cisco Packet Tracer. It has three guest floors, a reception area, an Accounts department and a central server room, all joined by one hierarchical network. Hotels now depend on ICT for guest Wi-Fi, bookings, staff communication, CCTV, door access and room automation, yet many still serve guests, staff and internal systems from the same unsegmented network. The result is congestion, weak security and no automation for lighting, climate, access or emergencies. This project replaces that with one segmented, routed and automated design.

Motivation

- **Public safety:** fire/smoke sensors trigger the alarm and sprinklers, RFID door locks control room entry, and motion-triggered CCTV covers the reception and hallway areas.
- **Societal / data protection:** guest traffic is kept away from the Accounts department so sensitive financial data stays confidential, while guests still get a working booking website and Wi-Fi.
- **Sustainability and cost:** lights, thermostats and air-conditioning respond to motion and room state through IoT rules instead of running constantly, reducing energy use and operating cost.
- **Regulatory / security posture:** the design follows the principle of least access: each group (guests, staff, IoT devices) reaches only the services it needs.

2. Topology

Relevant device(s):

Devices: five routers (labelled Router0, Router0(1), Router2, Router3 and Router4 in Packet Tracer), VLAN switches, wireless routers/access points, DNS, DHCP, Web, E-mail and IoT servers, staff PCs/laptops/printers, guest devices, and the IoT devices (RFID readers and cards, smart doors, motion detectors, webcams, thermostats, AC units, smart lights, fire/smoke sensors and sprinklers).

The topology is a **hybrid design: a hierarchical star inside each zone, joined by a ring of five routers**. Every router connects to its two neighbours over serial links (192.168.1.16/28, .48/28, .64/28, .96/28 and .128/28 in Fig 4.1.3) and uses a Gigabit LAN side to feed the switches of its zone. The zones are:

- **Guest floors:** each floor has its own switches and wireless routers/access points. Rooms contain IoT devices (RFID reader and door, light, thermostat, AC, fire sensor with sprinkler).
- **Reception:** staff PCs, a wireless router for lobby coverage, a webcam and a motion detector.
- **Accounts department:** a separate subnet behind its own router interface, so access to it can be controlled in one place.
- **Server room:** the central DNS, DHCP, Web, E-mail and IoT servers, kept in one /28 subnet so every zone reaches them through the routed core.

Why this arrangement: the ring gives each router two paths to the others, so OSPF can use an alternative route if a link fails. For example Router2 learns 192.168.1.128 over two equal-cost paths. Keeping each zone behind its own router interface means VLAN boundaries and ACLs line up with physical areas of the hotel. The IoT devices talk only to the IoT server, which keeps automation traffic separate from guest traffic.

3. Addressing Plan

Relevant zone(s)/subnet(s):

Subnets: 192.168.1.0/24 (split into /28 blocks), VLAN subnets 192.168.10.0/24 to 192.168.80.0/24, server subnet 192.168.1.192/28.

The whole network uses **private RFC 1918 addressing (192.168.0.0/16)** with a two-level VLSM plan. The routing table on every router reports "192.168.1.0/24 is variably subnetted, 19 subnets, 2 masks", which confirms that two prefix lengths are in use:

- **/28 blocks (255.255.255.240, 14 usable hosts)** carved out of 192.168.1.0/24. They are used for the router-to-router serial links, the router LAN-side segments, the Accounts department and the server room. Example: the link 192.168.1.16/28 joins Router0(1) (192.168.1.17) and Router4 (192.168.1.18). The server subnet 192.168.1.192/28 holds the Web server (192.168.1.196), the E-mail server (192.168.1.199) and the IoT server (192.168.1.200).

- **/24 subnets, one per VLAN (192.168.10.0 to 192.168.80.0):** these have more room for guest, staff and IoT devices. Each VLAN gateway is the .1 address on a router sub-interface (e.g. 192.168.10.1). Hosts receive addresses from the DHCP server; servers use static addresses.

Rationale and trade-off: small segments with few devices get small /28 blocks, while the VLANs that carry many wireless and IoT devices get full /24s. We used a /28 even for the point-to-point serial links so that the whole 192.168.1.x range follows one simple, uniform pattern that is easy to document and leaves room to grow. A /30 would use fewer addresses per link, which is the main efficiency trade-off of this choice.

IPv6: IPv6 was not implemented in this project. The whole design is IPv4 with private addressing, and dual-stack IPv6 is a possible later extension.

4. Routing

Relevant router(s):

Routers: all five routers run **OSPF (process 10)**.

OSPF was chosen because the hotel has several routers, floors and subnets that must reach each other. Static routes would need many hand-written entries on every router and would not adapt when a link fails; many of the simple projects reviewed in the literature stop at static or single-router routing. OSPF is a link-state protocol with fast convergence that scales as floors are added.

- **Adjacencies:** the console log of each router shows messages such as "%OSPF-5-ADJCHG: Process 10, Nbr 192.168.1.225 ... from LOADING to FULL", so neighbour relationships reach the FULL state on the serial links.
- **Learned routes:** "show ip route ospf" on Router2, Router3, Router4, Router0 and Router0(1) lists 19 subnets on every router, including all VLAN subnets. For example, Router2 reaches 192.168.10.0 via 192.168.1.49 and 192.168.60.0 via 192.168.1.66.
- **Costs:** path costs such as 65, 128, 129 and 192 are built from the default serial-link cost of 64 plus 1 for each Gigabit hop, and routes with equal cost are installed through two next hops (e.g. 192.168.1.128 on Router2).

OSPF also advertises the router-on-a-stick VLAN subnets, so a device in any VLAN can reach the servers and other zones without extra static routes.

5. VLANs and Inter-VLAN Routing

Relevant switch(es)/VLAN(s):

VLANs: 10, 20, 30 (Router0(1)); 40, 50 (Router4); 60, 70, 80 (Router0). Each VLAN is a separate /24 subnet. VLAN 40 is the guest VLAN used in the ACL test.

Hotel traffic is split into separate VLANs so that guest floors, reception/staff, the Accounts department and the IoT devices each have their own broadcast domain and subnet. This reduces broadcast traffic, stops guests from sharing a network with staff and internal systems, and lets access policy be applied per VLAN. The switches carry the tagged VLANs towards the routers.

Inter-VLAN routing uses **router-on-a-stick**: one Gigabit interface per router is divided into 802.1Q sub-interfaces (encapsulation dot1Q), each holding the gateway address of one VLAN. This keeps the router count low and puts every VLAN gateway where OSPF and the ACLs can act on it.

VLAN	Subnet	Gateway (sub-interface)	Router
10	192.168.10.0/24	192.168.10.1 (Gi0/0.10)	Router0(1)
20	192.168.20.0/24	192.168.20.1 (Gi0/0.20)	Router0(1)
30	192.168.30.0/24	192.168.30.1 (Gi0/0.30)	Router0(1)
40	192.168.40.0/24	192.168.40.1 (Gi0/0.40)	Router4
50	192.168.50.0/24	192.168.50.1 (Gi0/0.50)	Router4
60	192.168.60.0/24	192.168.60.1 (Gi0/0.60)	Router0
70	192.168.70.0/24	192.168.70.1 (Gi0/0.70)	Router0
80	192.168.80.0/24	192.168.80.1 (Gi0/0.80)	Router0

6. Security

Relevant device(s)/ACL/NAT config:

Devices: the router in front of the Accounts department (standard ACL), the IoT VLAN isolation, and the guest VLAN 40.
NAT: not configured (see below).

ACL policy. A standard ACL denies traffic whose source is the guest VLAN 40 subnet from reaching the Accounts department, and permits everything else. It is applied outbound on the router interface that faces the Accounts subnet, so the router checks every packet leaving towards Accounts. Because it filters only traffic bound for Accounts, guests keep access to the hotel web site and e-mail while the financial data stays protected. Staff devices can still reach the servers as required.

Why this design. Accounts holds the most sensitive data, so one ACL close to it gives strong protection with a very small rule set that is easy to audit. VLAN segmentation and the ACL together give layered security: VLANs separate the traffic, and the ACL enforces the policy between VLANs. IoT devices also communicate only with the IoT server, so a compromised guest device cannot reach them directly.

Evidence. In Fig 4.1.4, PC12 in the guest VLAN pings 192.168.1.82 in the Accounts subnet and all 4 packets fail ("Destination host unreachable", 100% loss).

NAT. No NAT configuration was implemented. The simulated hotel is a closed internal network that uses RFC 1918 private addresses throughout and has no ISP edge, so address translation was not needed. A real deployment would add NAT/PAT on an edge router to give guests and the hotel systems Internet access.

Not simulated: IDS/IPS, firewall appliances, network access control and encrypted IoT communication were beyond what Packet Tracer models; they are listed as future enhancements.

7. Wireless

Relevant device(s):

Devices: WRT300N wireless routers (for example "Wireless Router0" at reception), wireless access points in the guest rooms, and the guest laptops, smartphones and tablets that connect to them.

Wireless routers on each floor and at reception/lobby provide Wi-Fi coverage for guest rooms, the hotel lobby and staff areas. Wireless clients receive their IP configuration from the DHCP server and use the hotel web site through its server address (192.168.1.196), so they use the same addressing and services as wired devices.

Security choices. Each wireless router/AP has its own SSID for its floor or common area and is secured with WPA2. Beyond that, access is controlled at the network level: wireless guests sit in their own subnet and are subject to the same ACL policy, so they reach only permitted services and cannot reach the Accounts department. Enterprise-grade wireless security (WPA3-Enterprise authentication and network access control) was not simulated because of tool limitations and is listed as an improvement. The trade-off is coverage and ease of access for guests versus stricter authentication.

8. IoT / Smart-City Layer

Relevant device(s)/rule(s):

Devices: RFID readers and cards, smart doors, motion detectors, webcams (e.g. recep2cam), smart lights, thermostats, AC units, fire/smoke sensors, sprinklers and alarms, all registered on the IoT server at 192.168.1.200.

Every IoT device is registered on the IoT server (Fig 4.1.5 and Fig 4.2.1) and managed through its Conditions page. The IoT devices exchange data only with that server. Automation rules used in the model:

- **Motion to camera:** rule "rec2cameraON" turns the reception webcam ON when "recep2cammotion" is true, and "rec2cameraoff" turns it OFF when motion stops. Fig 4.2.1 shows the camera streaming once motion is detected.
- **RFID to door:** an RFID reader authenticates the card before the paired smart door opens. If a reader reports an invalid card (rule "r8dorinvalid": IoT28 Status is Invalid), the door IoT3 is set to Lock, as in Fig 4.2.2.
- **Light and climate:** the light and AC states drive the thermostats, for example "rm3ACon" sets room3ther to Cooling and "rm3ACoff" sets it to Heating; "rm2lightON/OFF" and "iot12lighton/off" set the matching thermostat the same way.
- **Fire response:** when the fire/smoke or temperature sensor detects abnormal heat, the IoT server activates the alarm and sprinkler.

Value. Safety: automatic fire response, controlled door access and motion-triggered CCTV. Sustainability: lights and climate react to room state instead of running all the time. Operations: less manual intervention for staff and better guest comfort. Note that Packet Tracer allows SMTP only from standard end devices, so IoT devices cannot send e-mail directly; the e-mail server serves staff PCs and the IoT server handles the automation.

9. Verification

Method(s) used (ping/traceroute/Simulation Mode):

Methods used: ping, IOS "show" commands on the router CLI, and application-level tests from client PCs (web browser, mail client, IoT server console).

Verification followed a layered order, from the lowest layer upwards:

- **Interfaces:** "show ip interface brief" on Router0(1), Router4 and Router0 confirms the Gigabit, sub-interface and serial links in use are up/up with the planned addresses, while unused interfaces stay administratively down.
- **Routing:** "show ip route ospf" on each router confirms that all subnets are learned and that adjacencies reach FULL (Fig 4.1.3).
- **Security:** a ping from guest PC12 to 192.168.1.82 fails with "Destination host unreachable", showing that the ACL works (Fig 4.1.4).
- **Services:** the web browser on PC12 loads the Premier Hotel site from 192.168.1.196; the mail clients send and receive messages through the POP3 server 192.168.1.199 ("Receive Mail Success"); and the IoT server console at 192.168.1.200 lists the registered devices with green status.
- **IoT automation:** motion detection switches the reception camera on and shows its image, and the RFID reader/door pair shows the card ID, status and lock state (Fig 4.2.1 and 4.2.2).

Testing in this order meant that a failure could be traced to the right layer: addressing and links first, then routing, then the ACL, then the services and IoT rules.

10. Supporting Screenshots

Topology Diagram:

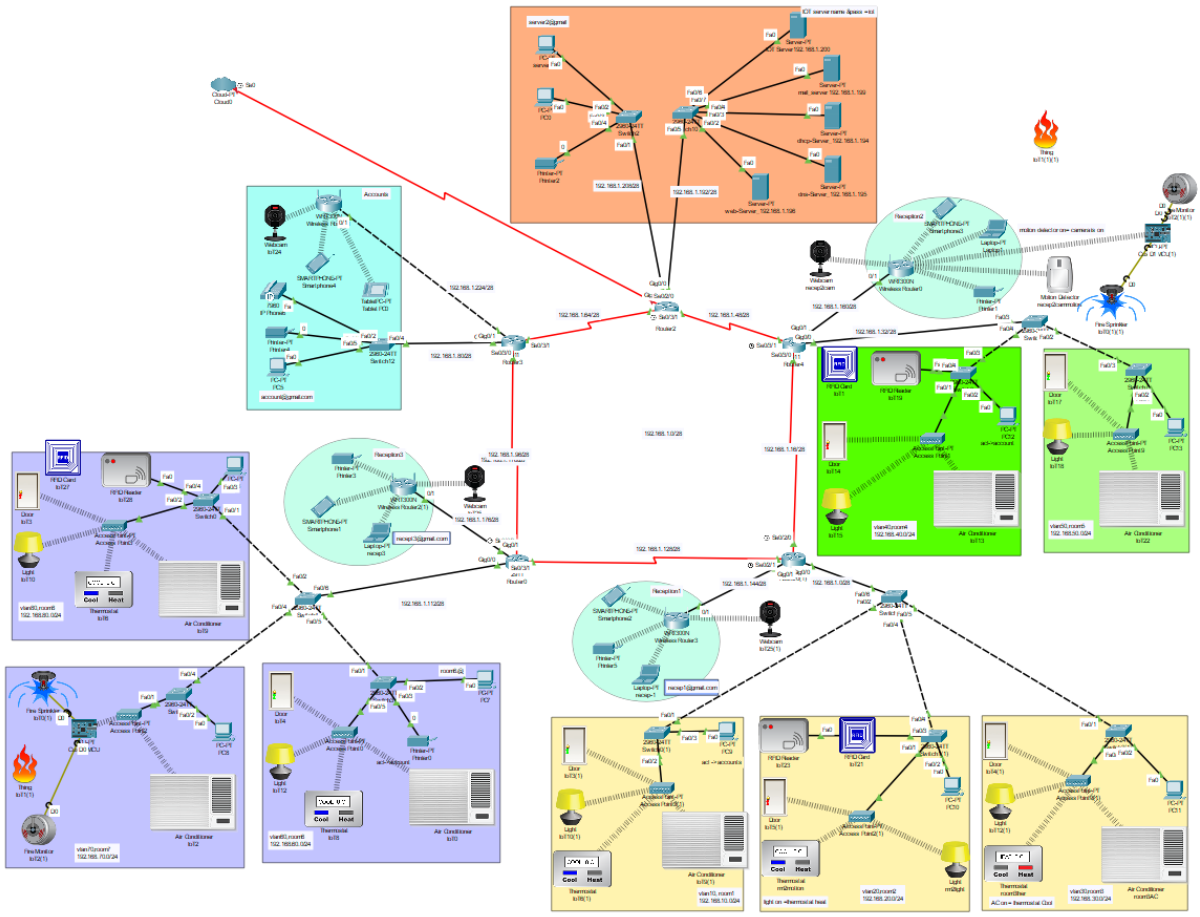


Fig 4.1.1: Overall Hotel Network Topology (five routers, switches, wireless devices, servers, guest floors, staff departments and IoT equipment)

IP Addressing Table:

Addresses are taken from the Router CLI output in Fig 4.1.2 (all 192.168.1.x interface addresses use /28 = 255.255.255.240; VLAN sub-interfaces use /24 = 255.255.255.0). Link subnets of Router2 and Router3 are shown in the OSPF routing tables in Fig 4.1.3.

Device	Interface	IP address	Prefix	Purpose
Router0(1)	Gi0/0	192.168.1.1	/28	LAN side / trunk base
Router0(1)	Gi0/0.10	192.168.10.1	/24	VLAN 10 gateway
Router0(1)	Gi0/0.20	192.168.20.1	/24	VLAN 20 gateway
Router0(1)	Gi0/0.30	192.168.30.1	/24	VLAN 30 gateway
Router0(1)	Gi0/1	192.168.1.145	/28	LAN segment
Router0(1)	Se0/2/0	192.168.1.17	/28	Serial link to Router4
Router0(1)	Se0/2/1	192.168.1.130	/28	Serial link to Router0
Router4	Gi0/0	192.168.1.33	/28	LAN side / trunk base
Router4	Gi0/0.40	192.168.40.1	/24	VLAN 40 gateway (guest)
Router4	Gi0/0.50	192.168.50.1	/24	VLAN 50 gateway
Router4	Gi0/1	192.168.1.161	/28	LAN segment
Router4	Se0/3/0	192.168.1.18	/28	Serial link to Router0(1)
Router4	Se0/3/1	192.168.1.49	/28	Serial link to Router2
Router0	Gi0/0	192.168.1.113	/28	LAN side / trunk base
Router0	Gi0/0.60	192.168.60.1	/24	VLAN 60 gateway
Router0	Gi0/0.70	192.168.70.1	/24	VLAN 70 gateway
Router0	Gi0/0.80	192.168.80.1	/24	VLAN 80 gateway
Router0	Gi0/1	192.168.1.177	/28	LAN segment
Router0	Se0/3/0	192.168.1.98	/28	Serial link to Router3
Router0	Se0/3/1	192.168.1.129	/28	Serial link to Router0(1)
Web server	NIC	192.168.1.196	/28	Hotel website (192.168.1.192/28)
E-mail server	NIC	192.168.1.199	/28	SMTP/POP3
IoT server	NIC	192.168.1.200	/28	IoT registration and rules

Device Configuration Screenshots:

VLAN sub-interface (router-on-a-stick) configuration and interface status:

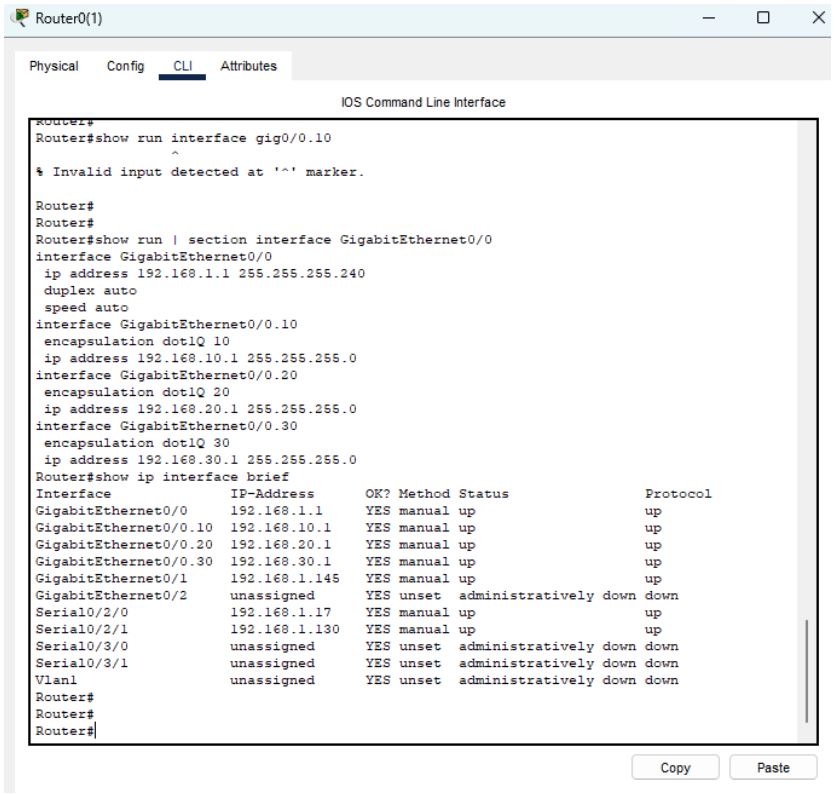


Fig 4.1.2 (a): Router0(1): VLAN 10/20/30 sub-interfaces and "show ip interface brief"

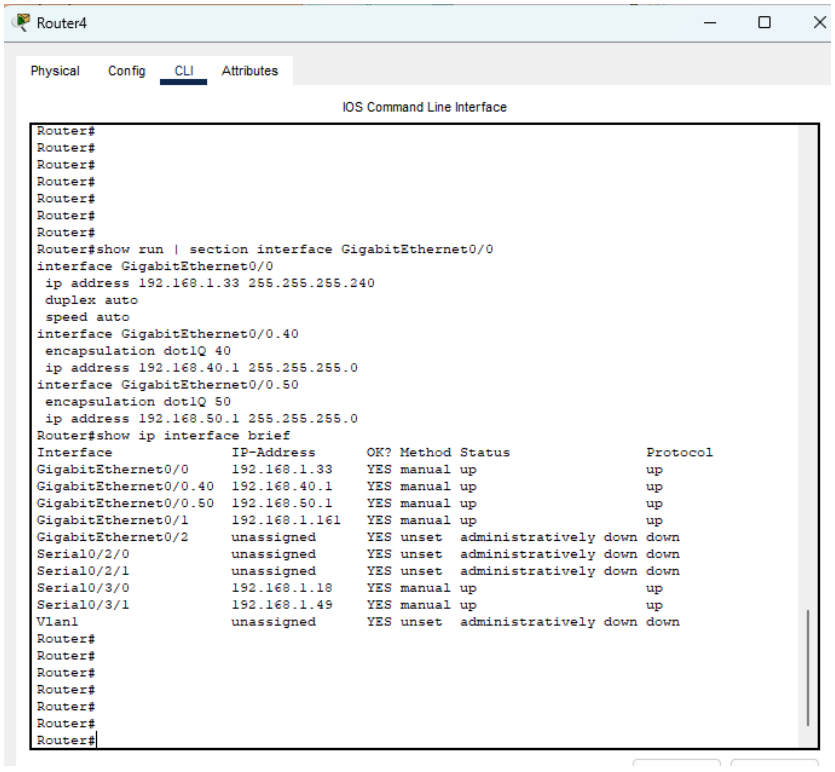


Fig 4.1.2 (b): Router4: VLAN 40/50 sub-interfaces and "show ip interface brief"

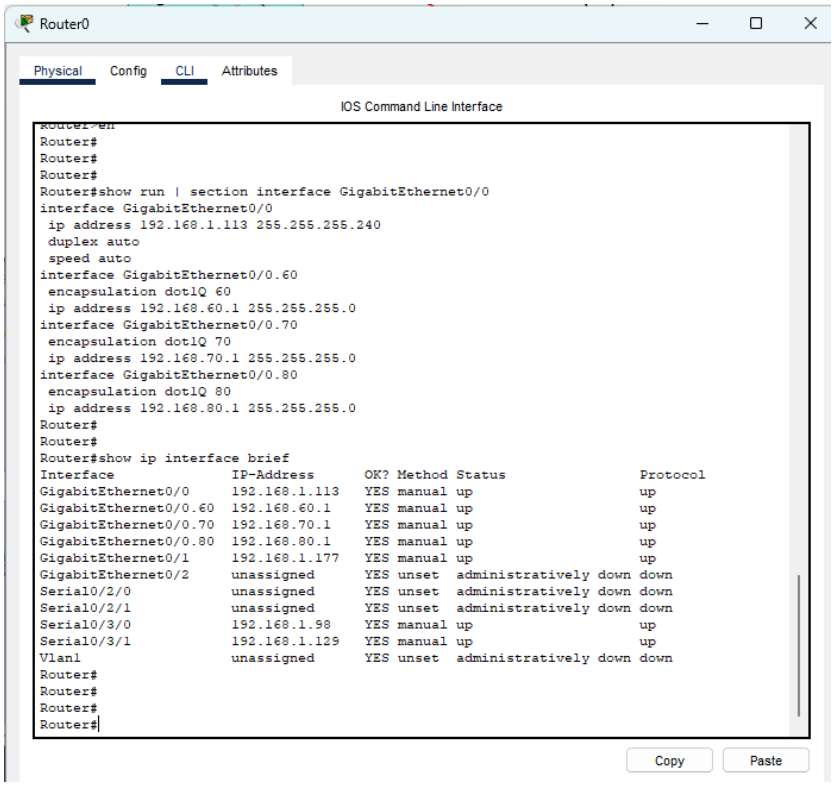


Fig 4.1.2 (c): Router0: VLAN 60/70/80 sub-interfaces and "show ip interface brief"

OSPF ring and routing tables:

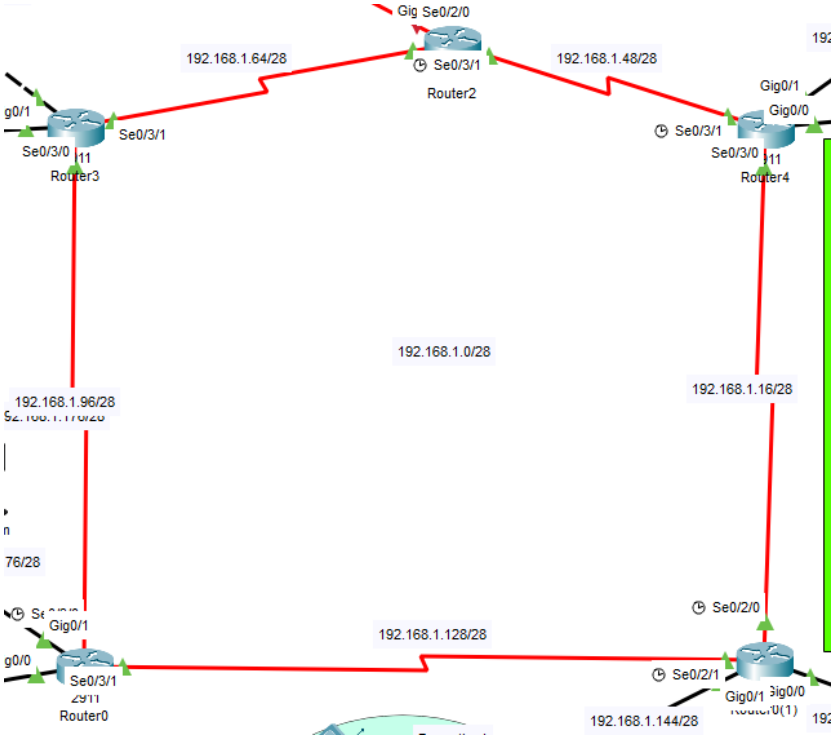


Fig 4.1.3 (a): Router ring with the serial-link subnets

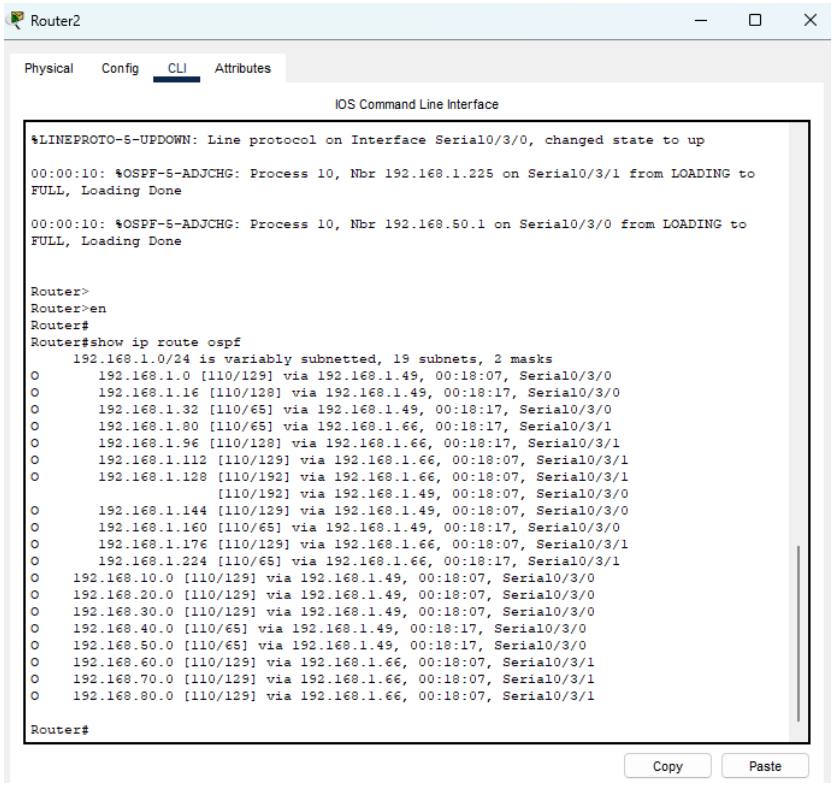


Fig 4.1.3 (b): Router2: OSPF adjacencies and "show ip route ospf"

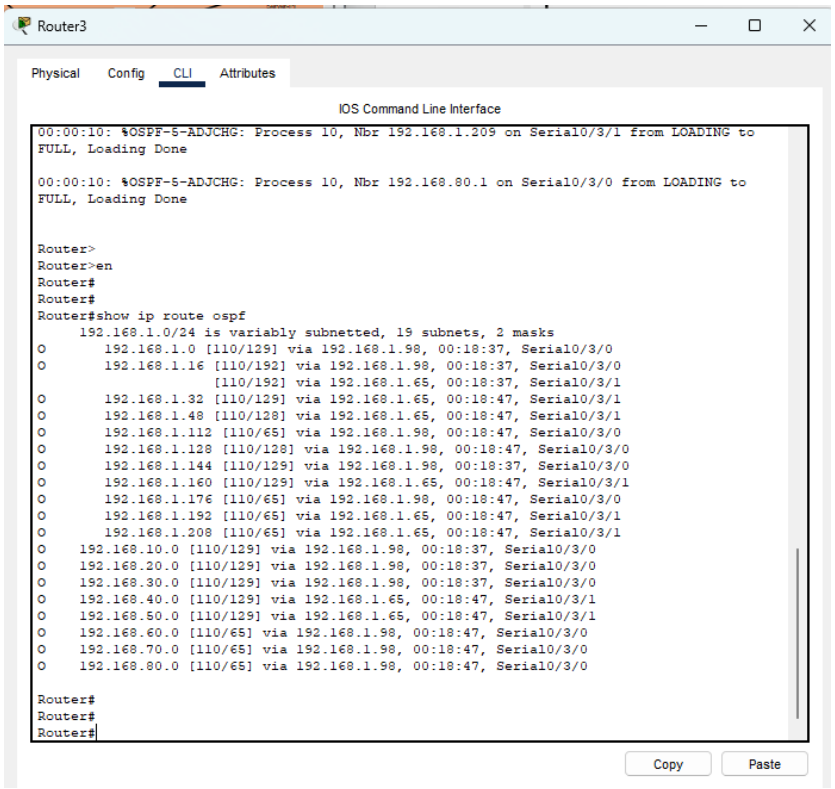


Fig 4.1.3 (c): Router3: OSPF adjacencies and "show ip route ospf"

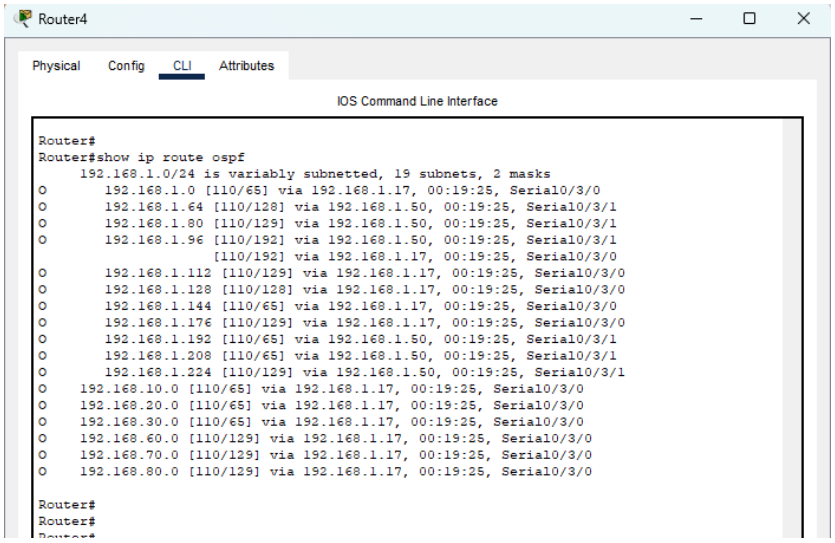


Fig 4.1.3 (d): Router4: "show ip route ospf"

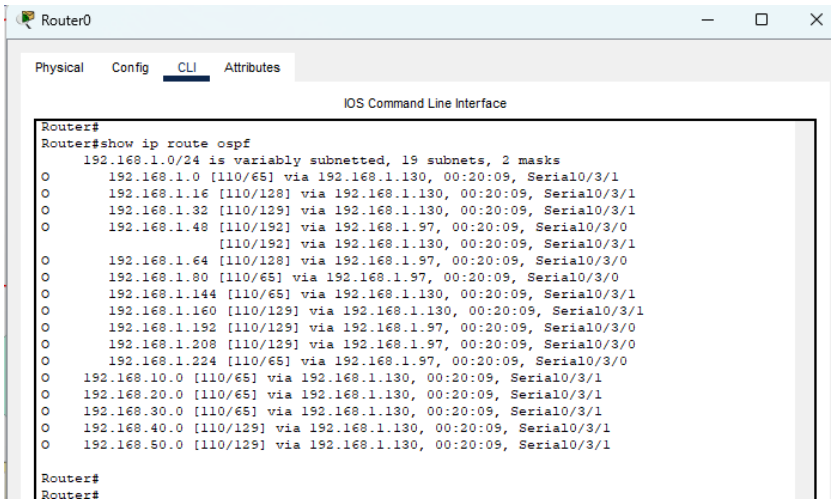


Fig 4.1.3 (e): Router0: "show ip route ospf"

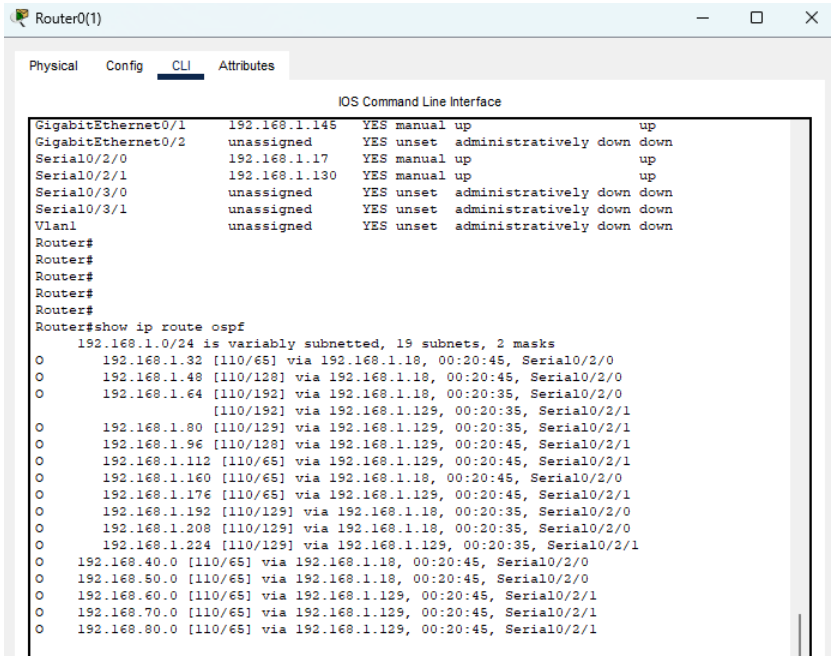


Fig 4.1.3 (f): Router0(1): interface status and "show ip route ospf"

Simulation Mode / Verification Screenshots (ping, traceroute, IoT automation triggering):

Security (ACL) test:

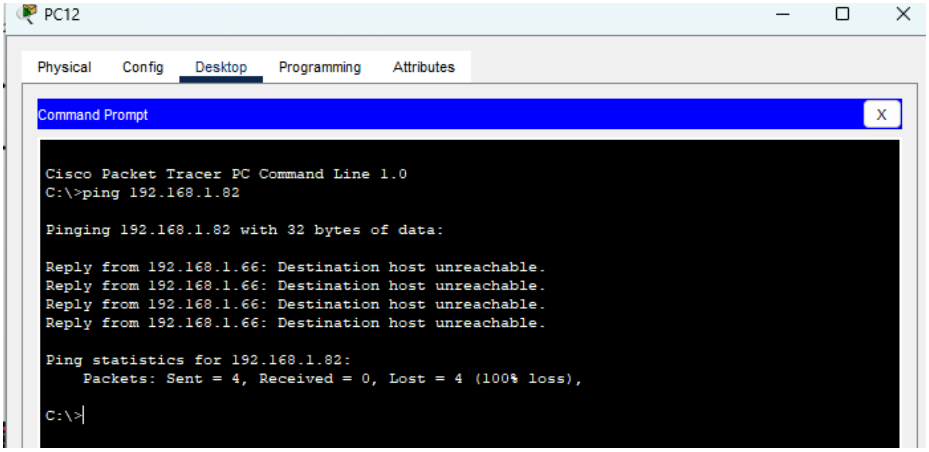


Fig 4.1.4: ACL blocking guest VLAN access to the Accounts department (ping from PC12 to 192.168.1.82, 100% loss)

Server tests: web, e-mail and IoT servers:

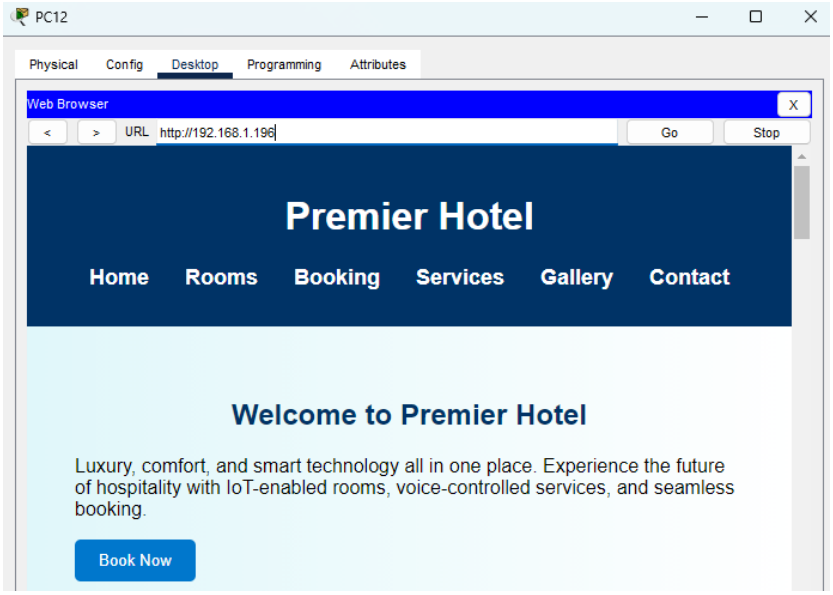


Fig 4.1.5 (a): Web server: Premier Hotel home page at http://192.168.1.196

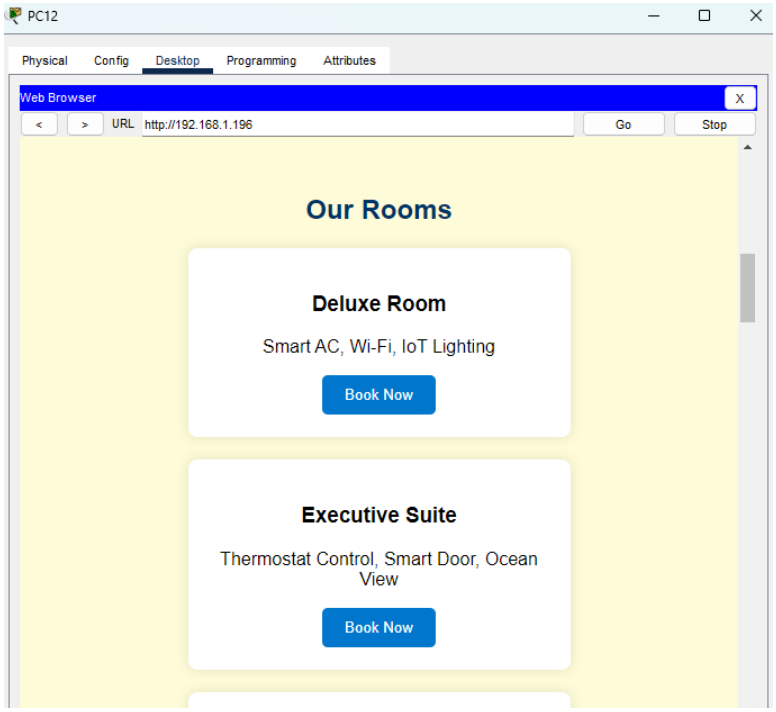


Fig 4.1.5 (b): Web server: "Our Rooms" page

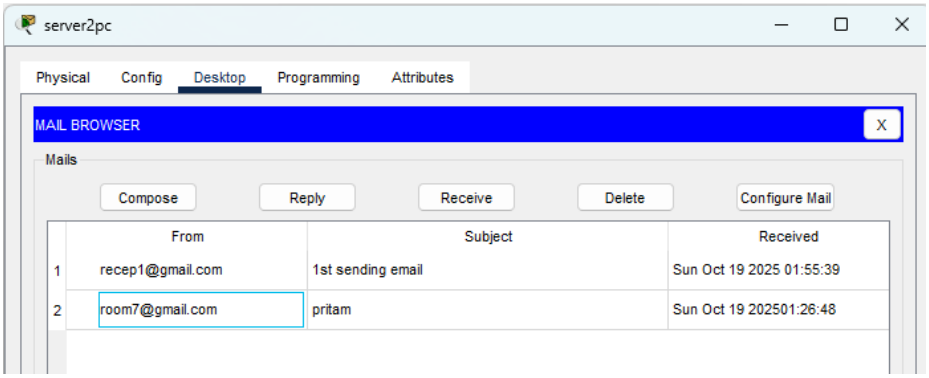


Fig 4.1.5 (c): E-mail server: mail received from staff PCs

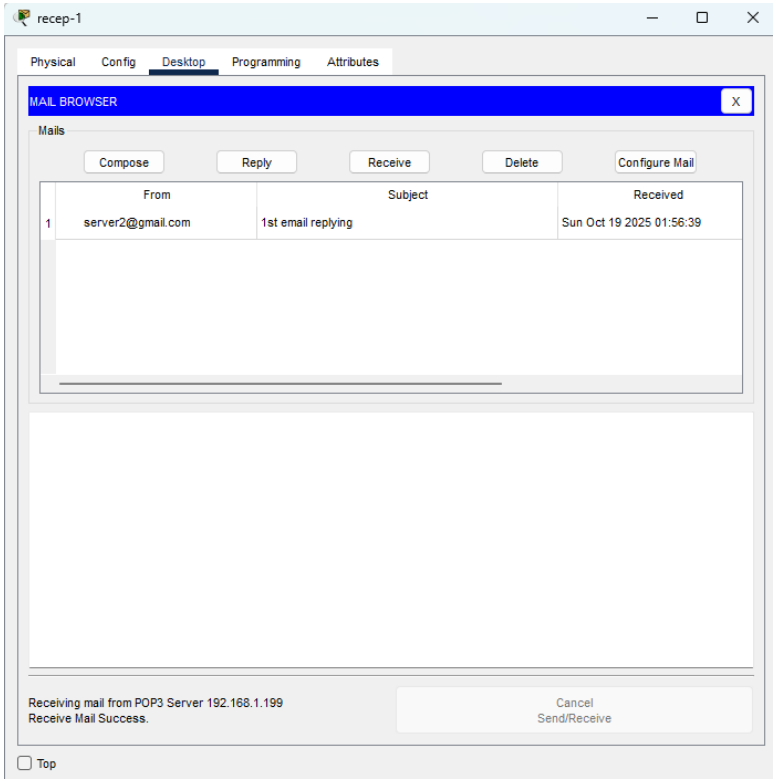


Fig 4.1.5 (d): Reception PC (recep-1): mail received from POP3 server 192.168.1.199

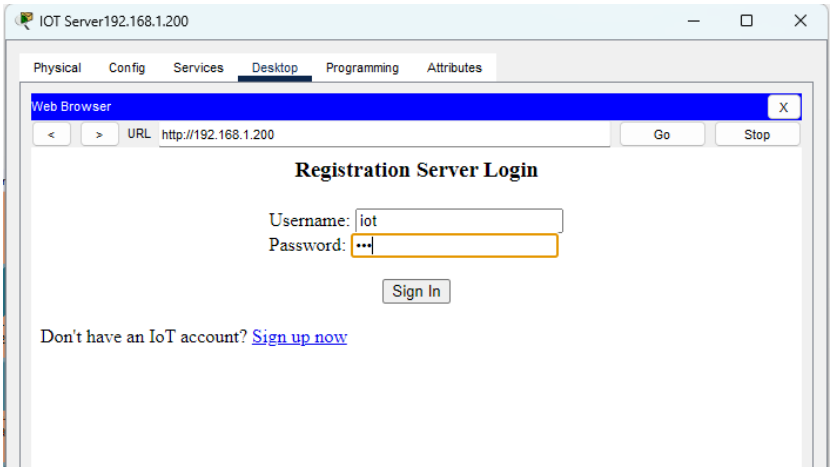


Fig 4.1.5 (e): IoT server login (192.168.1.200)

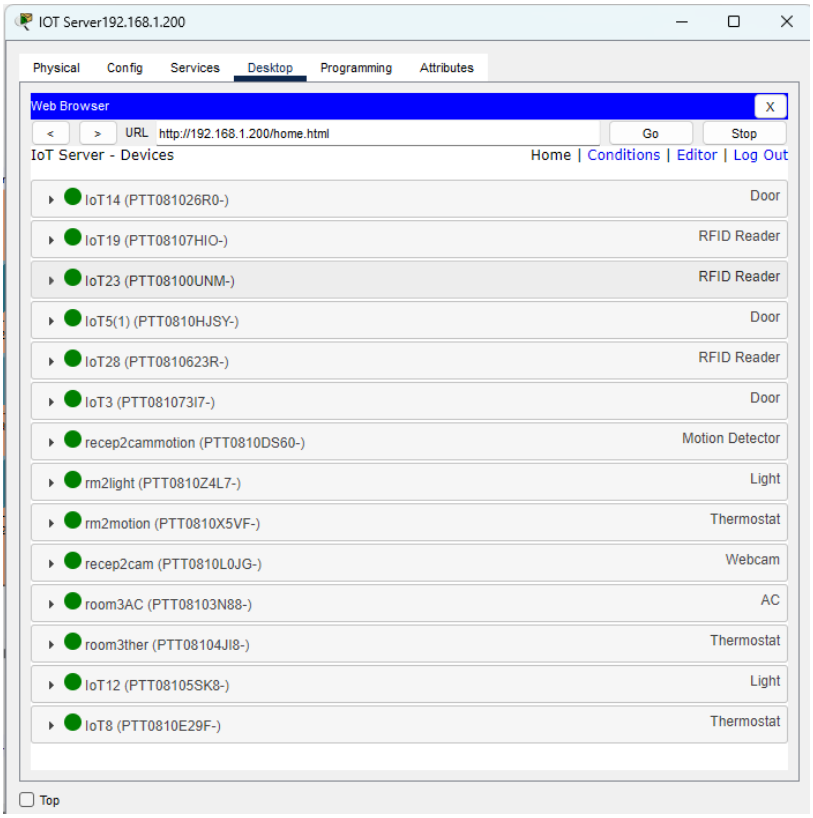


Fig 4.1.5 (f): IoT server: registered devices

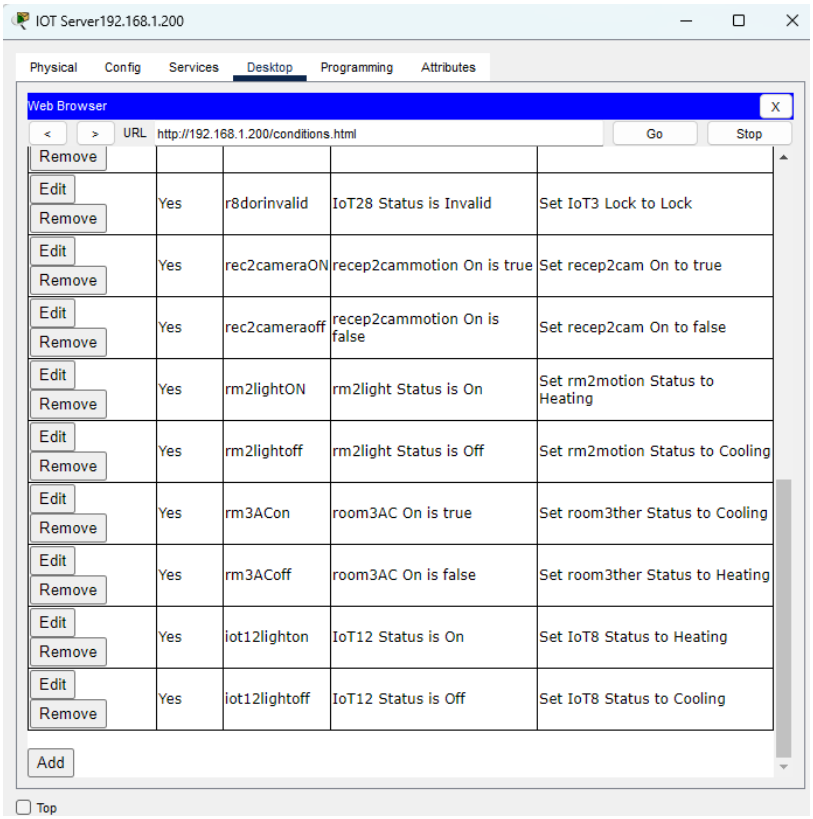


Fig 4.1.5 (g): IoT server: automation rules (conditions and actions)

IoT automation triggering:

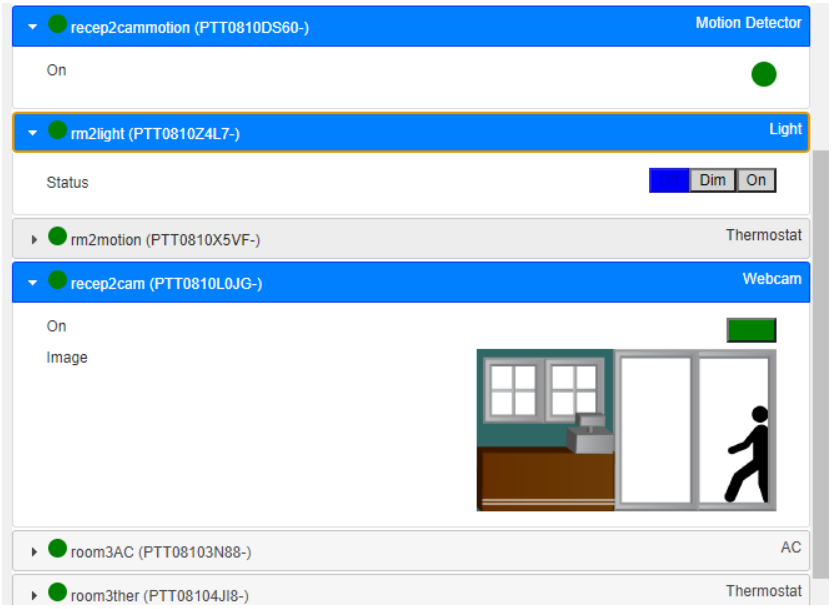


Fig 4.2.1: Motion detected, camera ON and image shown

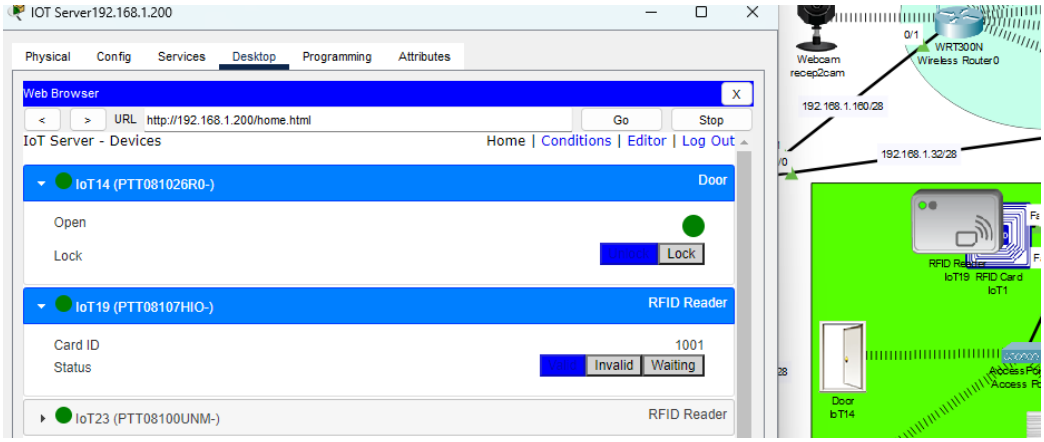


Fig 4.2.2: RFID access control for guest rooms

11. Knowledge Profile (KP) Reflection

#	Knowledge Area	Definition (BAETE) & Guiding Question	Your Response
K1	**Natural Sciences**	Theory-based understanding of the natural sciences relevant to the discipline. Often Not Applicable for a networking project — state clearly if no natural-science principles apply, or note if your IoT sensors model a physical quantity (e.g. light, temperature, motion).	Applies? ☐ Yes ☒ Partially ☐ No **How:** Partially applies. The IoT sensors model real physical quantities: motion for lighting and cameras, temperature and smoke for the fire response, and heating/cooling states for the thermostats and AC. No wider natural-science theory is used.
K2	**Mathematics**	Mathematics, numerical analysis, statistics, and formal computer/information science used for analysis and modelling. e.g. binary/subnet-mask arithmetic and VLSM calculations used to derive your addressing plan, or threshold values used in your IoT automation rule.	Applies? ☒ Yes ☐ Partially ☐ No **How:** Subnet-mask and binary arithmetic were used to divide 192.168.1.0/24 into /28 blocks (16 addresses, 14 usable hosts) and to give each VLAN its own /24. OSPF path costs in the routing tables (65, 128, 129, 192) are sums of the serial-link cost (64) and Gigabit cost (1), so cost arithmetic was used to understand route selection.
K3	**Engineering Fundamentals**	Theory-based engineering fundamentals underlying the discipline — for CSE, core networking/computing principles. e.g. the core networking fundamentals your whole design rests on: the OSI/TCP-IP layers, IP addressing, and switching/routing concepts.	Applies? ☒ Yes ☐ Partially ☐ No **How:** The design rests on the OSI/TCP-IP layers: VLANs and trunking at Layer 2, IP addressing and OSPF routing at Layer 3, and HTTP, SMTP/POP3, DNS and DHCP at the upper layers. Switching, routing and addressing fundamentals were applied together.
K4	**Specialist Engineering Knowledge**	Discipline-specific specialist knowledge and frameworks, including current practice areas — e.g. network design and routing. e.g. the specific techniques this course covers: VLSM/IPv6	Applies? ☒ Yes ☐ Partially ☐ No **How:** Applied VLSM addressing, OSPF dynamic routing, VLANs with router-on-a-stick inter-VLAN routing, standard ACLs, and IoT integration through an IoT server. NAT and IPv6 were not implemented.

#	Knowledge Area	Definition (BAETE) & Guiding Question	Your Response
		addressing, dynamic routing protocols (OSPF/EIGRP), VLANs and inter-VLAN routing, ACLs, NAT, and IoT integration.	
K5	**Engineering Design**	Knowledge that supports design in the practice area — e.g. network architecture and topology design. e.g. how you designed your zone layout, chose your addressing scheme, or decided where to place security boundaries.	Applies? ☒ Yes ☐ Partially ☐ No **How:** Designed a zone-based hierarchy (guest floors, reception, Accounts, server room) joined by a ring of five routers. Placed the server room centrally and put the ACL security boundary on the interface in front of Accounts, and chose the addressing scheme (/28 infrastructure blocks, /24 VLAN subnets) to match this layout.
K6	**Engineering Practice**	Knowledge of accepted engineering practice/technology in the field — e.g. simulation tools, CLI configuration, testing practices. e.g. your use of Cisco Packet Tracer, IOS CLI commands, consistent device naming, configuration documentation, or a verification routine you followed.	Applies? ☒ Yes ☐ Partially ☐ No **How:** Used Cisco Packet Tracer and IOS CLI commands ("show run | section interface", "show ip interface brief", "show ip route ospf") to build and verify the network, used descriptive device names for the IoT devices (recep2cam, rm2light, room3AC), and followed a layered verification routine (interfaces, routing, ACL, services, IoT).
K7	**Engineering and Society**	Understanding of engineering's role in society: ethics, professional responsibility, and social/economic/environmental impact. e.g. how your ACL policy or IoT automation protects public-safety or resident data, or why your smart city/town/region scenario's real-world context matters to its users.	Applies? ☒ Yes ☐ Partially ☐ No **How:** The ACL protects the hotel's financial data from guest access, the fire sensors, alarms, sprinklers and RFID locks support guest safety and security, and motion/temperature automation reduces energy use. These are real concerns for hotel guests, staff and owners.
K8	**Research Literature**	Engagement with research literature or authoritative technical references in the discipline. e.g. Cisco's official documentation, RFCs (e.g. on VLSM or private addressing), or any other technical reference you consulted beyond the lecture notes.	Applies? ☐ Yes ☒ Partially ☐ No **How:** Cisco reference documentation for OSPF and VLAN configuration was consulted, together with four references beyond the lecture notes: a SlideShare network-topology presentation (https://www.slideshare.net/siddiqueibrahim37/networktopologyppt), a Creately hotel IT network diagram (https://creately.com/diagram/example/h1vycihg1/Hotel IT Network%250Plan), the Princeton COS 461 topology slides (https://www.cs.princeton.edu/courses/archive/spring06/cos461/slides/13Topology.ppt) and an open-source GitHub hotel network project (https://github.com/Waqar-Ali010/Hotel-Network-Project). RFCs were not formally cited.

12. Complex Engineering Problem (CEP) Reflection

#	Attribute	Definition (BAETE) & Guiding Question	Your Response
P1	**Depth of Knowledge Required**	Resolved using in-depth, fundamentals-based engineering/specialist knowledge — not solved by routine or superficial application of a single taught example. e.g. does your design require you to genuinely apply networking fundamentals — addressing, routing, VLAN segmentation, security — in an integrated way, rather than each in isolation?	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. Addressing, routing, VLAN segmentation, security and services had to be applied together rather than separately. For example, an IoT device needs the right VLAN subnet, a DHCP address, an OSPF route across several routers and an ACL that does not block it before it can reach the IoT server.
P2	**Range of Conflicting Requirements**	Involves wide-ranging or conflicting technical and other issues that must be balanced against each other. e.g. does your scenario force trade-offs — addressing efficiency vs. scalability, security restrictiveness vs. usability, wireless coverage vs. security?	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. The design balances guest access against protection of Accounts (the ACL blocks Accounts but keeps web and e-mail open), IoT isolation against the need to reach the IoT server, and uniform /28 addressing against address efficiency, all within what Packet Tracer can simulate.
P3	**Depth of Analysis Required**	Has no single obvious solution; requires abstract thinking and originality to formulate a suitable design. e.g. did you have to weigh multiple possible topologies, addressing schemes, or routing protocols before settling on one, rather than there being one "correct" answer?	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. There was no single obvious design. We weighed segmenting the hotel per floor versus per function, and interconnecting the five routers as a full mesh versus a hierarchical star with OSPF, before settling on a hierarchical design with per-zone VLAN segmentation. OSPF was also preferred over the static or single-router routing typical in earlier projects.
P4	**Familiarity of Issues**	Involves issues that are infrequently encountered, rather than routine, well-rehearsed problems. e.g. does your ACL policy or IoT automation rule address a scenario-specific condition that is not simply the standard lab-manual example?	Addressed? ☐ Yes ☒ Partially ☐ Not Applicable **Justification:** Partially. Hotel-specific conditions go beyond a basic lab example: RFID invalid-card lock-out, motion-triggered CCTV, and linked light/AC/thermostat rules. The ACL itself is a standard one.
P5	**Extent of Applicable Codes**	Goes beyond standard codes of practice — typically most relevant to safety-critical or regulated engineering domains. Often Not Applicable at this project scale — state clearly if no formal codes of practice apply to your scenario, or note any addressing/security standard you chose to follow voluntarily (e.g. RFC 1918 private addressing).	Addressed? ☐ Yes ☐ Partially ☒ Not Applicable **Justification:** Not applicable: no formal code of practice or regulation applies at this scale. RFC 1918 private addressing was followed voluntarily for all addresses.
P6	**Extent of Stakeholder Involvement**	Involves diverse groups of stakeholders with widely varying needs and requirements. e.g. does your smart city/town/region scenario need to serve residents, public-safety staff, and network administrators, each with different access needs?	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. The hotel serves guests (Wi-Fi, booking site), reception and Accounts staff, managers and IT administrators, and the IoT devices themselves, each needing different access: guests reach only permitted services, Accounts is restricted, and IoT devices reach only the IoT server.

#	Attribute	Definition (BAETE) & Guiding Question	Your Response
P7	**Interdependence**	Is a high-level problem made up of many interdependent component parts or sub-problems. e.g. do your routing, VLANs, addressing, security, wireless, and IoT layers all have to work together correctly for the network to function — a change in one affecting the others?	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. Each layer depends on the others: VLAN to subnet to router sub-interface to OSPF advertisement to ACL to DHCP/DNS to IoT server reachability. A wrong gateway or subnet breaks routing, which breaks service access and IoT automation.

13. Complex Engineering Activity (CEA) Reflection

#	Attribute	Definition (BAETE) & Guiding Question	Your Response
A1	**Range of Resources**	Involves the use of diverse resources — people, time, information, and technologies. e.g. Cisco Packet Tracer, IOS command references, instructor consultation during the Lab 11 design session, or peer discussion used during development.	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. Resources used were Cisco Packet Tracer, IOS CLI command references, the course lab manual, four online references, the hotel website pages hosted on the web server, and the IoT server with its device registration and rules, plus the time spent on design, testing and reporting.
A2	**Level of Interaction**	Requires resolving significant problems arising from interactions between wide-ranging or conflicting technical issues. e.g. resolving a conflict between your VLAN/addressing plan and your routing configuration, or between your ACL policy and legitimate IoT/wireless traffic.	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. Interactions had to be resolved between the ACL and legitimate traffic (Accounts blocked but web and e-mail kept open), between IoT isolation and server access, and between the SMTP restriction of IoT devices in Packet Tracer and the planned e-mail alerts.
A3	**Innovation**	Involves creative use of engineering principles and knowledge in novel ways. e.g. the originality of your smart city/town/region scenario, or a design choice (addressing scheme, routing protocol, IoT automation rule) you arrived at rather than copied from a lab example.	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. The smart hotel scenario combines a multi-page hotel website (Home, Rooms, Booking, Services, Gallery, Contact), RFID access, motion-triggered CCTV and linked light, thermostat and AC rules in one design. The automation rules were written for the rooms and reception of this hotel.
A4	**Consequences for Society and the Environment**	Has significant consequences in a range of contexts, where predicting and mitigating those consequences is difficult. e.g. your scenario's public-safety, societal, or sustainability motivation (Lab 4/13's smart-city theme), or a safety/reliability consideration in your network design.	Addressed? ☒ Yes ☐ Partially ☐ Not Applicable **Justification:** Yes. Fire detection and sprinklers, door access control and CCTV address guest safety; ACL isolation protects financial data; and automated lighting and climate control reduce energy use and operating cost.
A5	**Familiarity**	Extends beyond previous experience, requiring you to apply principles-based approaches to a new or unfamiliar situation. e.g. does your scenario or a specific technique (e.g. IPv6 addressing, WLC configuration, IoT scripting) go beyond what was directly demonstrated in the lab manual?	Addressed? ☐ Yes ☒ Partially ☐ Not Applicable **Justification:** Partially. The IoT server registration and rule scripting went beyond the basic lab exercises. IPv6 and wireless LAN controller (WLC) configuration were not attempted.

14. Challenges Encountered and How They Were Resolved

Problem Encountered	How You Resolved It
Keeping guest traffic away from the Accounts department. Without separation, unrestricted communication could expose sensitive administrative data.	Placed the networks in separate VLANs and added a standard ACL on the router interface facing the Accounts subnet that denies guest VLAN 40. A ping from PC12 to 192.168.1.82 now fails with "Destination host unreachable", while guests still reach the web and e-mail services.
Making five routers and eight VLAN subnets reach each other reliably.	Used OSPF (process 10) on all routers and router-on-a-stick sub-interfaces for the VLANs. "show ip route ospf" shows all 19 subnets on each router and adjacencies at FULL, so every VLAN can reach the servers.
Packet Tracer allows SMTP only from standard end devices, so IoT devices (for example fire sensors) cannot send e-mail alerts directly.	Limited the e-mail server to staff PCs and moved all IoT automation (alarm, sprinkler, camera, door lock) onto the IoT server rules. The e-mail server is used for staff communication, as shown in Fig 4.1.5.
Getting the many IoT devices to respond together (door, camera, light, thermostat, AC).	Registered every device on the IoT server (192.168.1.200), gave each a clear name (e.g. recep2cam, rm2light, room3AC) and wrote conditions that link one device state to another. All devices show a green status and the motion to camera and RFID to door rules trigger as expected.

15. Conclusion

This project designed and simulated the network of **Premier Hotel**, a multi-floor smart hotel, in Cisco Packet Tracer. It combines VLSM addressing, eight VLANs with router-on-a-stick inter-VLAN routing, OSPF across five routers, a standard ACL that protects the Accounts department, wireless access, and centralised DHCP, DNS, web, e-mail and IoT servers. Verification showed that the routers form FULL OSPF adjacencies and learn every subnet, that guest traffic cannot reach Accounts, that the hotel website and mail service work, and that IoT automation (motion to camera, RFID to door lock, light and climate rules) responds correctly.

The design shows that routing, segmentation, security, servers and IoT only work when they are planned together: a change in the addressing plan, for instance, affects the VLAN gateways, the OSPF routes, the ACL and the IoT server reachability. Along the way we learned how to divide an address block with VLSM, how router-on-a-stick and OSPF work together, how to place an ACL close to the resource it protects, and how Packet Tracer limits what the IoT devices can do (for example, SMTP from IoT devices).

Limitations and future work. NAT, IPv6, IDS/IPS, WPA3-Enterprise/NAC and router redundancy (HSRP/VRRP) were not implemented. Future improvements are stronger security, backup routers, cloud services for booking and storage, smarter IoT automation (AI-based energy control, voice-activated settings, predictive maintenance) and a central monitoring system using SNMP or dashboards.

Declaration

I declare that this report and the accompanying source code / configuration are our own original work, prepared for this course, and that all external references or assistance used have been acknowledged.

Student Signatures: Sazib Hossain: _____ Mohammad Rayan: _____ Ifty Anwar: _____ Santusti Biswas: _____ Date:	Student IDs: 0222320005101174 (Sazib Hossain) 0222320005101186 (Mohammad Rayan) 0222320005101192 (Ifty Anwar) 0222320005101194 (Santusti Biswas)

Appendix: Device Configuration Scripts

Interface configuration of the routers (from "show run | section interface GigabitEthernet0/0", Fig 4.1.2):

Router0(1)

```
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.240
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
```

Router4

```
interface GigabitEthernet0/0
 ip address 192.168.1.33 255.255.255.240
 duplex auto
 speed auto
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 192.168.50.1 255.255.255.0
```

Router0

```
interface GigabitEthernet0/0
 ip address 192.168.1.113 255.255.255.240
 duplex auto
 speed auto
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 192.168.60.1 255.255.255.0
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 192.168.70.1 255.255.255.0
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 192.168.80.1 255.255.255.0
```